

Andrew Chen

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PROFESSIONAL SUMMARY

Statistics and Economics student at UC Davis specializing in machine learning and AI systems. Experienced in building deep learning models in PyTorch and designing LLM-powered agent. Strong foundation in statistics, optimization, and algorithms, with hands-on experience developing end-to-end ML pipelines, NLP models, and scalable data workflows. Fast learner who is passionate about translating complex data into intelligent systems that are robust, interpretable, and production-ready.

EDUCATION

University of California, Davis

B.S. in Statistics & B.A. in Economics | Expected Graduation: May 2027

- Cumulative GPA: 3.94 / 4.00
- Awards: Dean’s Honor List in Fall 2023, Winter 2024, Spring 2024, Spring 2025
- Certificate: IBM Data Science Certificate, Macquarie University Excel Skills for Business

PROFESSIONAL EXPERIENCE

Research Assistant

UC Davis Department of Economics | Sep 2025 – Present

- Collected and digitized 50 years of weekly central bank gold reserve data from The Economist for a historical macroeconomics project studying international monetary transmission during the pre-WWII Gold Standard era.
- Designed an optimization algorithm leveraging each issue’s 3-week coverage pattern, reducing required downloads to 34.6% while preserving full data integrity.
- Automated the end-to-end pipeline using Selenium for large-scale PDF scraping and Python tools such as pdf2image and Gemini API to convert scanned documents into structured CSV datasets.

Student Assistant

UC Davis Technology Transfer Office | Aug 2025 – Present

- Manage weekly intake and initial processing of new Record of Invention submissions, ensuring complete documentation and smooth entry into the university’s technology commercialization pipeline.
- Verify sponsor information, MTAs, and supporting documents with high accuracy to maintain compliance and prepare cases for patent evaluation.
- Organize and maintain sensitive legal and technical files for the university’s patent portfolio, ensuring accuracy and confidentiality for all case files.

Operational Management Intern

PAGO LLC | Aug 2025 – Sep 2025

- Analyzed large operational datasets using SQL and Excel to uncover performance trends, automate reporting, and support data-driven decision-making.
- Built a centralized tracking system for 5,000+ SKUs and worker KPIs, streamlining data pipelines and reducing manual processing by 5 hours per week.
- Recommended a revised workflow based on data analysis, which was implemented and led to a 20% reduction in average order processing time.

PROJECTS

Agentic Stock Copilot

- Designed and deployed an end-to-end agentic stock copilot integrating the OpenAI API, Alpaca market data API, and AWS cloud storage to retrieve, store, and analyze real-time financial data for beginner investors.
- Engineered deterministic financial analysis functions for trend, momentum, and volatility indicators, preventing LLM hallucination and enabling a hybrid AI system that grounds LLM outputs in mathematically computed market signals.
- Developed a structured LLM orchestration workflow where natural language queries are parsed into validated function calls using Pydantic schemas, enforcing strict type constraints and ensuring robust execution of financial computations.

Fake Review Detection

- Architected and trained LSTM and transformer-based BERT classification models in PyTorch, implementing gradient optimization, attention masking and fine-tuning strategies to achieve 97% accuracy, outperforming classical ML model.
- Conducted feature engineering and linguistic analysis using text preprocessing, TF-IDF, embedding, and tokenization to quantify stylistic and semantic differences between machine-generated and human-written text.
- Built a production-ready text classification pipeline with rigorous evaluation using precision, recall, F1 score and confusion matrix, delivering a scalable model capable of detecting sophisticated synthetic content.

SKILLS

- Programming:** Proficient in Python and R. Experienced in algorithm design, data pipeline development, web scraping/automation, building end-to-end workflows, and developing web interfaces using HTML/CSS.
- Machine Learning:** Supervised/unsupervised learning (SVM, Random Forest, PCA, clustering); deep learning in PyTorch (CNNs, LSTMs, Transformers); model optimization, regularization, and evaluation (ROC-AUC, F1, MAE).
- AI:** NLP (TF-IDF, word tokenization and embedding), LLM-based system design with function calling, hybrid deterministic-computation pipelines, API integration, and schema validation/type enforcement using Pydantic.
- Data Querying & Analysis:** SQL (JOINS, CTEs, window functions); Excel (pivot tables, VLOOKUP).
- Data Visualization:** matplotlib, seaborn, plotly, ggplot2; Tableau for dashboards and interactive data storytelling.
- Soft Skills:** Self-driven and proactive learner who adapts quickly to new environments; Strong communicator who collaborates effectively across teams; Highly organized with excellent time-management and attention to detail.